

Child Marriage Bans and Female Schooling and Labor Market Outcomes: Evidence from Natural Experiments in 17 Low- and Middle-Income Countries[†]

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I measure the effect of child marriage bans on female educational attainment and employment using a difference-in-differences approach employing subnational spatial and cohort variation in a sample of over 250,000 female respondents from 17 low- and middle-income countries banning child marriage between 1995 and 2012. My results using the full study sample suggest that raising the minimum legal age of marriage to 18 increased age at marriage, age at first birth, and the likelihood of employment. In urban areas, the bans also reduced child marriage and increased educational attainment. Effects of the bans typically were smaller in rural areas, in countries with a lower pre-ban minimum legal age at marriage, for cohorts with lower temporal exposure to the ban, and for reducing marriage at ages just below 18, consistent with imperfect enforcement. (JEL I21, J12, J13, J16, J18, K38, O15)

Early marriage is associated with poor socioeconomic and female empowerment outcomes among women and low overall levels of economic development and growth. Recent quasi-experimental research offers insights on how to reduce early marriage (e.g., Duflo et al. 2015; Buchmann et al. 2017; Chow and Vivalt 2022), yet there is little research on a common—albeit not universal—policy: child marriage bans. Evidence on whether these bans affect early marriage and whether these changes affect female schooling and labor market outcomes may offer insights into how to address this issue at scale. In this paper, I measure the effects of child marriage bans on female schooling and labor market outcomes.

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Between 1995 and 2012, more than 25 low- and middle-income countries banned female child marriage.¹ I use these legislative changes as a set of natural experiments to identify the effects of child marriage bans on female schooling and labor market outcomes. I employ a difference-in-differences type specification using individual-level data and exploiting subnational geographic and country-specific cohort variation to measure the effects of the bans. Following the empirical strategy employed in settings with similar natural experiments (e.g., Bleakley 2003, 2007, 2010; Bleakley and Lange 2009; Cutler et al. 2010; Lucas 2010, 2013; Feyrer et al. 2017), I compare the change in child marriage associated with the timing of the child marriage ban in subnational regions where the pre-ban age at marriage was low to the synchronous change subnational regions where the pre-ban age at marriage was high. This approach controls for time-invariant heterogeneity across subnational regions and for country-specific cohort effects. For an omitted factor to explain my findings, it must be a factor that changed at the same time as the legislation in each country and changed differentially in subnational regions with high/low pre-ban ages at marriage in each country. To assess the plausibility of my identification assumption, I present evidence from event study analyses of the five main sets of outcomes—married before age 18, age at marriage, age at first birth, years of schooling, and employment—as well as a host of auxiliary outcomes.

Data on legislative changes in the minimum legal age of marriage come from the MACHEquity Child Marriage Policy Database (World Policy Center 2017), spanning the period 1995–2012. Data on age at marriage, fertility timing, education, and employment come from the Demographic and Health Surveys (DHS), national household surveys that offer standardized sociodemographic and employment data across countries. The 17 countries that enacted legislative changes during the period 1995–2012 banning child marriage and for which there is a postban DHS constitute the study sample.²

My results using the full study sample indicate that raising the minimum legal age of marriage to 18 increased age at marriage, age at first birth, and the likelihood of employment. In urban areas, the bans also reduced child marriage and increased educational attainment. For example, I find that a one standard deviation increase in ban intensity was associated with an approximately 0.6 year increase in age at marriage, an approximately 0.4 year increase in age at first birth. These are fairly modest reductions/increases in relative terms: approximately 3 percent and 2 percent, respectively, compared to the sample means. I also find that a one standard deviation increase in ban intensity increased employment by approximately 7.2 percentage points (approximately 13 percent relative to the sample mean).³ In the urban sample, I find roughly equivalent effects on these outcomes as in the full sample.

¹The United Nations defines a child as an individual under the age of 18 (United Nations 1989).

²My main analysis pools respondents from all 17 countries, while allowing for all control parameters to be country-specific. I show that the main results are not driven by any single country and that the general pattern of findings applies across subgroups of countries. Sample size limitations (i.e., fewer than 30 subnational units) preclude me from reliably estimating country-specific effects of the bans rather than the overall, pooled effect and effects for large subgroups of respondents and countries.

³It is also possible to scale effect sizes in terms of the representative country. Consider the urban-only sample. Median intensity in this sample is 0.57. My main results for this sample suggest that for the median urban area, the ban would have reduced child marriage by approximately 5.2 percentage points and increased age at marriage and

In addition, in the urban sample I find that a one standard deviation increase in ban intensity reduced child marriage by 4.5 percentage points (approximately 20 percent compared to the sample mean) and increased years of schooling by 0.52 years (approximately 6 percent compared to the sample mean).

In general, the effects of the bans appear to be less pronounced in rural areas. For example, the bans do not appear to have reduced child marriage *per se* in rural areas. Yet my results indicate meaningful effects on age at marriage (with particularly large reductions in the likelihood of marriage at younger ages (e.g., age 13 or 14)), years married before age 18, and age at first birth, consistent with the hypothesis of a greater degree of imperfect enforcement in rural areas. In contrast, the bans appear to have had a greater effect on employment in rural areas than in urban areas, without increasing years of schooling. Supply-side heterogeneity and/or systematic differences in the transition to adulthood in rural versus urban areas may explain this seemingly anomalous finding.

Consistent with a causal interpretation of these results, I find that they are robust to alternative control cohorts, alternative treatment cohorts, alternative measures of subnational regional intensity, and a variety of individual, spatial, and temporal controls, including subnational regional time trends. Event study analyses indicate the absence of differential pre-ban trends, reveal sharp changes in outcomes around the timing of the ban, and suggest larger effects among younger respondents among the subset of cohorts exposed to the bans. Instrumental variables estimates using the bans as an instrument for age at marriage indicate that among urban respondents postponing marriage by one year increased educational attainment by approximately 0.4 years. In the full sample (and in rural areas), the IV regressions suggest a “null effect” of postponing marriage on educational attainment.⁴

My findings contribute to several bodies of economic literature. First, a broad body of economic literature has examined joint determinants of female marriage, fertility timing, schooling, and labor market outcomes. Much of this literature has focused on the roles of labor market opportunities (e.g., Jensen 2012; Shah and Steinberg 2017), household structure (e.g., Vogl 2013), or technological changes such as “the pill” (e.g., Goldin and Katz 2000, 2002; Bailey 2006, Bailey 2010; Edlund and Machado 2015). I complement this literature by demonstrating that legislation—a seemingly accessible policy lever—can have large, simultaneous effects on these outcomes as well.

Second, a handful of economic studies have used data from the United States to examine the effect of child marriage laws, or child marriage laws bundled with other laws, on marriage and schooling outcomes or have used these laws as instruments

age at first birth by approximately 0.77 and 0.60 years, respectively. It would have increased years of schooling by approximately 0.61 years, and increased employment by approximately 3.0 percentage points.

⁴I discuss concerns about the exclusion restriction in Section IVD.

for age at marriage in poverty or wage regressions.^{5,6,7} Dahl (2010) used a combination of legal changes, including laws such as minimum age at marriage laws and compulsory schooling laws, as instrumental variables for age at marriage in poverty regressions, finding that younger marriage caused increased poverty. Bharadwaj (2015) examined the effect of a 1957 amendment to a Mississippi marriage law that increased the minimum legal age at marriage (for females and males), strengthened parental consent requirements, imposed compulsory blood tests, and required proof of age, finding that the bundle of legislative changes reduced marriage and fertility, while increasing educational attainment. Wang and Wang (2017) examined the effect of minimum age at marriage laws in the United States on wages, finding that these laws increased wages by increasing educational attainment. I expand on these studies by: (i) providing evidence from outside of the United States; (ii) providing evidence from multiple natural experiments conducted at different times and in different locations; (iii) isolating the effect of child marriage laws from other laws; and (iv) simultaneously presenting evidence on marriage, fertility, schooling, and labor market outcomes.

Third, there appears to be one other economic study of the effects of child marriage laws in low- and middle-income countries, written in parallel to the current study. Collin and Talbot (2017) also combined the MACHEquity Child Marriage Policy Database and the DHS to examine the effects of child marriage laws on child marriage, finding little to no evidence of effects. Three key features distinguish my analysis from this study. Perhaps most importantly, I employ a cross-sectional source of variation other than birth cohort: the subnational region pre-ban intensity of child marriage. In contrast, Collin and Talbot (2017) employed a regression discontinuity (RD) approach testing for discontinuities at age 18 (and other minimum legal ages) in the distribution of age at marriage, relying on cohort variation within countries.⁸ To the extent that the laws were not immediately and perfectly enforced, the RD design is likely to underestimate the effects of these bans.⁹ In addition, I define my study sample as respondents in countries experiencing a child marriage ban, whereas Collin and Talbot (2017) includes over 60 countries, more than 40 of which did not experience a child marriage ban. Finally, I present evidence on a broad set of economic outcomes, including fertility, schooling, and labor market outcomes, not just age at marriage.

⁵Several economic studies also examine the effects of changes in other marriage laws on age at marriage and marriage rates, such as lowering the age of marital nonconsent from 21 to 18 (Blank et al. 2009) and changes in blood test requirements for marriage (Buckles et al. 2011).

⁶In broadly related economic literature on the effects of marriage bans on human capital outcomes, Francis et al. (2011) uses number of ballot measures as an instrumental variable to measure the effects of (the effects of lifting) same-sex marriage bans in the United States on risky sexual behavior and sexually transmitted infections.

⁷A few recent economic studies also provide evidence from field experiments in poor countries on the efficacy of economic incentives and educational programming designed to reduce early marriage (e.g., Duflo et al. 2015; Buchmann et al. 2017; Chow and Vivalti 2022).

⁸Collin and Talbot (2017) also examine minimum legal ages without parental consent, whereas, as I discuss in Section IIIA, I focus on the minimum legal age with parental consent because legal child marriage without parental consent is much less common in these data.

⁹My event study analyses (e.g., Figures 4–8) suggest that the child marriage bans had little effect on girls age 17 at the time of the ban, yet typically large and statistically significant effects for younger girls.

Fourth, several economic studies measure the effect of age at marriage on schooling and other outcomes (e.g., Field and Ambrus 2008; Hicks and Hicks 2015; Chari et al. 2017). This literature largely has used age at menarche as an instrumental variable for age at marriage. Although age at menarche is a strong predictor of age at first marriage, it may not satisfy the exclusion restriction for many sets of outcomes. Males who value female empowerment outcomes (e.g., female employment) will likely prefer to marry females who are older than the youngest brides. In dynamic models of marriage markets, for any set of biologically eligible females from a given birth cohort, the highest match-quality females will be “fished out” at the younger eligible ages, with the pool of unmatched females from that birth cohort decreasing over time due to compositional changes. Thus, males who value female empowerment will select on older females from a given birth cohort who have only more recently become biologically eligible. This selection dynamic creates a positive association between age at menarche and empowerment outcomes (e.g., female employment) that is not driven by age at marriage. My study provides new evidence on the causal effect of age at marriage using a different instrumental variable that plausibly does not directly affect female schooling and labor market outcomes. Moreover, the magnitude of estimate of the causal effect of age at marriage on educational attainment among urban respondents in my analysis—delaying marriage by one year caused 0.4 years of additional schooling—is approximately the same as in this literature, supporting the external validity of both sets of studies.¹⁰

The rest of the paper is organized as follows. Section I provides background on child marriage bans in international law. Section II describes the empirical strategy. Section III describes the data. Section IV presents the results. Section V concludes.

I. Background on Child Marriage Bans in International Law

The United Nations’ 1948 Universal Declaration of Human Rights appears to provide the modern international legal basis for banning child marriage. Article 16 of the Declaration identifies two key legal provisions: (i) men and women of full age are entitled to equal rights as to marriage; and (ii) marriage requires free and full consent of both spouses (United Nations 1948).

Three subsequent international agreements have strengthened this legal foundation. The United Nations’ 1962 Convention on Consent to Marriage, Minimum Age for Marriage, and Registration of Marriages requires that all parties enact a legal minimum age for marriage and register all marriages (United Nations 1962). Articles 15 and 16 of the United Nations’ 1979 Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) require parties to provide women and men with equal rights regarding freedom in choosing their “residence and domicile” and in choosing their spouse, respectively (United Nations 1979).

¹⁰ Ambrus and Field (2008) found in Bangladesh that delaying marriage by one year increased educational attainment by 0.22 years. Hicks and Hicks (2015) found in Kenya that delaying marriage by one year increased educational attainment by 0.48 years. Chari et al. (2017) examined the effects on children’s human capital outcomes, not the reference female’s human capital.

The United Nations' 1989 Convention on the Rights of the Child formalized the definition of a child as anyone under the age of 18 (United Nations 1989).

These agreements belie mixed formal support by countries around the world. For example, only 61 of the 193 member states of the United Nations participate in the 1962 Convention on Consent to Marriage, Minimum Age for Marriage, and Registration of Marriages (United Nations 2019a). Similarly, although 191 member states participate in the 1979 Convention on the Elimination of All Forms of Discrimination Against Women, many participant member states have Declarations and/or Reservations about minimum age at marriage stipulations, which typically mean that the countries do not observe the United Nations' standard that the legal minimum age of marriage be age 18 for females (United Nations 2019b).¹¹

Despite this seemingly overall moderate level of formal support for these agreements, a substantial fraction of countries around the world have banned child marriage in the past three decades. The 1995 Beijing Declaration and Platform for Action called on countries to "enact and strictly enforce" a legal minimum age at marriage and reaffirmed age 18 as the accepted legal minimum age at marriage, in effect calling on countries to ban child marriage (United Nations 1995). After the Beijing Declaration, many countries passed child marriage bans. In 1995, estimates suggested that 80 percent of countries worldwide allowed child marriage (Arthur et al. 2014).¹² By 2013, this number had fallen to 56 percent (Arthur et al. 2014).¹³

Although many countries banned child marriage after the Beijing Declaration, this did not mean that these bans were perfectly enforced. Qualitative and quantitative evidence suggest imperfect enforcement, particularly in rural areas. For example, the United States Department of State (2015a) notes that Albania did not effectively enforce their 2003 child marriage ban and that "underage marriages occurred mostly in rural areas and within Romani communities." Likewise, the United States Department of State (2015b) notes that Ethiopia did not uniformly enforce their 2000 child marriage ban and that awareness of the law was low in rural areas. Wodon et al. (2017) estimate that at least 7.5 million girls were married illegally in 2017.

II. Empirical Strategy

A. Regression Specification

My empirical strategy employs the interaction of pre-ban heterogeneity in the typical age at marriage with the timing of the ban in a difference-in-differences type approach. This strategy closely follows the approach in other settings with similar natural experiments (e.g., Bleakley 2003, 2007, 2010; Bleakley and Lange 2009;

¹¹ As another example, the United States is a signatory on all of these agreements, yet at least 46 U.S. states appear to allow child marriage (Cornell Law School 2019).

¹² As described in this sub-section, the United Nations categorizes all marriage under the age of 18 as child marriage, invariant of parental consent. My analysis uses this terminology and it guides the focus of the empirical analysis on legislative changes banning marriage before age 18.

¹³ For comparison, estimates suggest that 23 percent of countries worldwide allowed child marriage without parental consent in 1995 and that this number fell to 12 percent by 2013 (Arthur et al. 2014).

Cutler et al. 2010; Lucas 2010, 2013; Feyrer et al. 2017). My primary regression equation is

$$(1) \quad outcome_{icrj} = \beta_0 + \beta_1 intensity_{rj} * post_{cj} + \gamma_{cj} + \delta_{rj} + \varepsilon_{icrj},$$

where $outcome_{icrj}$ is the outcome of interest (e.g., years of schooling) of respondent i in cohort c in subnational region r in country j . The regressor $intensity_{rj}$ is a measure of the pre-ban intensity of child marriage for a given country-region-urban combination. The regressor $post_{cj}$ is an indicator variable equal to one if the respondent was under the age of 18 at the time of the legislative change and zero otherwise. The terms γ_{cj} and δ_{rj} are country-specific cohort effects and subnational region fixed effects,¹⁴ respectively, and ε_{icrj} is an idiosyncratic error term. I estimate the parameters of equation (1) using ordinary least squares (OLS) regression and estimate heteroskedasticity-robust standard errors clustered at the subnational region (i.e., country-region-urban combination) level.^{15, 16}

Throughout the regression analysis, I estimate the parameters of equation (1) for three separate samples: the full study sample, rural areas, and urban areas. This allows for the estimated effect of the ban to vary systematically with rural-urban residence, the primary observable dimension on which enforcement varies according to the contextual analysis in Section I.

I interpret β_1 as the causal effect of the bans on the outcome of interest. Controlling for country-specific cohort fixed effects addresses concerns about country-specific life-cycle differences in marriage, human capital investment decisions, and labor market outcomes. Controlling for subnational region fixed effects addresses concerns about time-invariant unobserved heterogeneity in these outcomes. In addition, the subnational region fixed effects capture the fact that the regressor $intensity_{rj}$ likely is mechanically associated with the age at marriage outcome (i.e., individuals in subnational regions where child marriage is common are likely to have married at a young age). For an omitted variable to bias the estimates of β_1 , the omitted variable must vary systematically by the subnational region level and this correlation must have changed systematically with the timing of the bans. To illuminate the plausibility of this assumption, I complement my difference-in-differences type regressions with event-study analyses testing for placebo “effects” and displaying the dynamic effects of the bans. I also conduct a variety of robustness and sensitivity checks, including controlling for subnational regional trends.

¹⁴ My main specification defines the subnational region as the interaction of DHS region and urban.

¹⁵ There are 282 country-region-urban combinations in my study sample (virtually equally divided between urban and rural areas), suggesting that standard asymptotic tests are appropriate (Cameron et al. 2008).

¹⁶ I omit the typical notation for the vector of individual-level controls. My main specification does not include any individual-level controls and I examine robustness to including individual-level controls (e.g., country-specific religion), the parameters of which I allow to vary across countries. Using a main specification that relies on regional and cohort variation in exposure to the timing of the ban helps avoid controlling for a mechanism by which the bans may have affected an outcome of interest.

B. Measuring Ban Intensity

My main measure of subnational regional ban intensity treats the ban as having greater intensity in locations where the pre-ban average age at marriage was lower. Consider a location where prior to the ban one-half of females married at age 17 and one-half married at age 18. This would result in an average age of marriage of 17.5, or 0.5 years under age 18. Under universal compliance, the ban would raise the average age of marriage in this location by 0.5 years. In contrast, consider a location where prior to the ban one-half of females married at age 16 and one-half married at age 17, resulting in an average age of marriage of 16.5, or 1.5 years under age 18. Under universal compliance, the ban would raise the average age of marriage in this location by 1.5 years. To model this effect, I define the main measure of ban intensity for subnational region r in country j as follows:

$$(2) \quad intensity_main_{rj} = \sum_{i=1}^N \frac{I(\text{married before age 18})_{irj} * (18 - \text{marriage age}_{irj})}{cohort\ size_{rj}},$$

where $intensity_main_{rj}$ is the summation across females in the pre-ban cohorts. The numerator is defined as zero for respondents remaining unmarried by age 18. The denominator $cohort\ size_{rj}$ is the number of females in the pre-ban cohorts. This main ban intensity measure captures pre-ban heterogeneity in child marriage along two dimensions: (i) the extensive margin (i.e., likelihood married before age 18); and (ii) the intensive margin (i.e., 18 minus marriage age, for individuals marrying before age 18).

The alternative measure of subnational-regional ban intensity treats the ban as having had greater intensity in locations where the pre-ban prevalence of child marriage was greater. This measure captures variation in the pre-ban intensity of child marriage on the extensive margin and, unlike the main intensity measure, ignores variation in the pre-ban intensity of child marriage on the intensive margin. The alternative intensity measure is defined as

$$(3) \quad intensity_alternative_{rj} = \sum_{i=1}^N \frac{I(\text{married before age 18})_{irj}}{cohort\ size_{rj}},$$

where $I(\text{married before age 18})_{irj}$ is an indicator variable equal to one if individual i married before age 18. As with the main measure of ban intensity, this alternative measure is defined using pre-ban cohorts. The alternative intensity measure provides a useful robustness check to help ensure that the construction of the main intensity measure is not driving the results (e.g., an average measure may be sensitive to outliers), although—unlike the main intensity measure—the alternative measure does not capture the distance by which a particular region falls short of having no child marriage.

III. Data

A. Legislative Changes in Minimum Legal Age of Marriage

Data on legislative changes in minimum legal age of marriage come from the MACHEquity Child Marriage Policy Database (World Policy Center 2017).

Researchers at the McGill Institute for Health and Social Policy and the UCLA Fielding School of Public Health collected these data in 2013–2014 using funding from the Bill and Melinda Gates Foundation, the Bill, Hillary, and Chelsea Clinton Foundation, and the William and Flora Hewlett Foundation. This panel dataset spans the period 1995–2012 and includes information on minimum legal age with parental consent and minimum legal age without parental consent.¹⁷ Minimum legal age was determined by consulting country government Websites, the Lexadin World Law Guide, the Foreign Law Guide, the NATLEX database, and other sources (MACHEquity 2014).¹⁸ It was recorded as “18 years or older,” “16 or 17 years old,” “14 or 15 years old,” “9 to 13 years old,” or “No minimum age,” separately for females and for males.¹⁹ The legal age at marriage is lower for females than males in many (if not most) countries, the average age at marriage is lower for females than for males, and the literature on the effects of age at marriage primarily studies female age at marriage. Thus, I focus on changes in the legal minimum age at marriage for females. I exclude countries for whom data are not available at the start of the panel (i.e., 1995), resulting in a balanced panel of 105 countries.

Figure 1 displays the number of countries allowing child marriage (for females) by year for the entire period 1995–2012, disaggregated by parental consent. Legally allowing child marriage without parental consent was not common in these countries. At the start of the panel, around 80 percent of these countries allowed child marriage with parental consent. By the end of the panel, nearly one-third of those countries had banned child marriage with parental consent. Therefore, the regression analysis examines the effects legislative changes in the minimum age at marriage with parental consent.

The next step in sample construction is to exclude countries that did not allow female child marriage as of 1995. The MACHEquity dataset does not include information on legal minimum age at marriage prior to 1995, meaning it does not allow for the identification of the year of a legislative change for child marriage bans implemented prior to 1996. I discuss the final step in my sample construction in the following subsection.

B. *Individual-level Outcomes*

Individual-level data on demographic and socioeconomic outcomes come from the Demographic Health Surveys (DHS) (Demographic and Health Surveys 2017). The DHS are national household surveys conducted in over 90 low- and middle-income countries. Among the 105 countries in the MACHEquity data, there are 60 countries that have at least one DHS survey round. Of these countries, 17 completely banned

¹⁷ In instances where the minimum legal age varies across subnational regions (e.g., in federal systems), these data code the lowest minimum legal age in a country as the country’s minimum legal marriage age.

¹⁸ MACHEquity (2014) states: “If legislation was not available, analysts reviewed the most recent reports submitted by countries to the monitoring committees of the Convention on the Rights of the Child (CRC) and the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), as well as the reports detailing the committees’ concluding observations.”

¹⁹ A majority of countries around the world allow exceptions to these age limits under specific conditions: pregnancy, having given birth to a child, or a court or government office granting permission (Arthur et al. 2014). Depending on the prevalence of these conditions, they may attenuate the potential effects of child marriage bans.

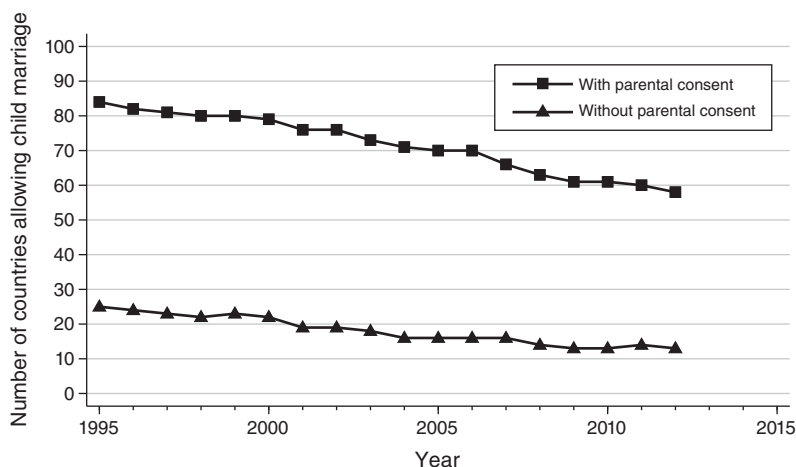


FIGURE 1. NUMBER OF COUNTRIES ALLOWING CHILD MARRIAGE

Notes: Data come from the WORLD Policy Analysis Center. Sample restricted to those countries where 1995 data are available.

marriage under the age of 18 (i.e., invariant of parental consent), forming my study sample.^{20,21} Table 1 displays the set of countries that comprise my primary sample. Approximately 40 percent of sample countries had a pre-ban legal minimum of age 14 or 15. Another approximately 40 percent of sample countries had a pre-ban legal minimum of age 16 or 17. Maldives, Nigeria, and Sierra Leone had no minimum legal age prior to banning child marriage.

Table 2 displays summary statistics for the DHS female respondents in the sample countries. The average age is around 30 years old, 42 percent of the sample is urban, and around 60 percent of respondents were married at the time of the survey. More than one-third of respondents were married before age 18, yet only 6 percent were married before the age of 14. Among ever married respondents, the average age of first marriage was nearly 19 years old. A quarter of respondents reported primary schooling as the highest level of schooling attended. Approximately one-half of respondents were currently employed, the majority of whom were employed for cash and were employed all year.²² The urban sample is older, married at older ages, received more years of schooling, and was less likely to be employed than the rural sample.

Figure 2 displays the joint timing of the survey rounds and the ban years. Panel A reveals a broadly even range of ban years, with the median respondent experiencing

²⁰Overlap between the DHS and countries that banned child marriage without parental consent during the period 1995–2012 is substantially smaller than for bans of child marriage with parental consent. In addition, the countries that banned child marriage without parental consent during this period and are present in the DHS are the countries that banned child marriage with parental consent. These countries are Albania, Benin, Democratic Republic of the Congo, Egypt, Ethiopia, Jordan, Kazakhstan, and Maldives.

²¹I use data from the most recent DHS in each country as of May 2017.

²²The Jordan DHS and Kazakhstan DHS did not ask female respondents about compensation type. The Jordan DHS did not ask respondents about employment timing (e.g., all year, seasonal).

TABLE 1—COUNTRIES, SURVEY YEARS, MINIMUM LEGAL AGE AT MARRIAGE, AND KEY SUMMARY STATISTICS BY COUNTRY

		Minimum legal age at marriage with parental consent:			Interval between ban and interview		
Country (1)	Ban year (2)	Year = 1995 (3)	Year = 2012 (4)	Interview years (5)	year (6)		
Panel A. Ban years and interview years by country							
Albania	2003	16 or 17	18 or older	2008/9	5		
Benin	2004	14 or 15	18 or older	2011/12	7		
Dem. Republic of the Congo	2009	14 or 15	18 or older	2013/14	4		
Egypt	2008	16 or 17	18 or older	2014	6		
Ethiopia	2000	14 or 15	18 or older	2003	3		
Guinea	2008	16 or 17	18 or older	2012	4		
Jordan	2001	14 or 15	18 or older	2012	11		
Kazakhstan	1998	16 or 17	18 or older	1999	1		
Liberia	2012	16 or 17	18 or older	2013	1		
Madagascar	2007	14 or 15	18 or older	2008/9	1		
Maldives	2001	No minimum	18 or older	2009	8		
Namibia	1996	14 or 15	18 or older	2013	17		
Nepal	2003	16 or 17	18 or older	2011/12	8		
Nigeria	2003	No minimum	18 or older	2013	10		
Peru	1999	14 or 15	18 or older	2000	1		
Sierra Leone	2007	No minimum	18 or older	2013	6		
Togo	2007	16 or 17	18 or older	2013/14	6		
				Married before age 18	Age at first birth	Education	Employed
Country (1)	Observations (2)	Age (3)	Married (4)	age 18 (5)	(6)	(7)	(8)
Panel B. Summary statistics by country							
Albania	7,584	31.63	0.64	0.08	22.70	10.11	0.30
Benin	16,599	28.97	0.56	0.32	9.81	2.74	0.64
Dem. Republic of the Congo	18,811	28.29	0.49	0.39	19.20	5.71	0.68
Egypt	21,755	33.03	0.94	0.26	21.63	8.33	0.16
Ethiopia	16,508	27.70	0.57	0.47	18.79	3.29	0.36
Guinea	9,138	28.59	0.73	0.53	18.20	2.55	0.72
Jordan	11,352	34.43	0.95	0.21	22.30	10.94	0.18
Kazakhstan	4,799	31.29	0.60	0.08	21.97	11.09	0.42
Liberia	9,227	29.16	0.33	0.40	18.15	3.47	0.55
Madagascar	17,340	28.74	0.60	0.41	19.22	4.33	0.81
Maldives	7,036	32.55	0.92	0.36	20.25	5.93	0.42
Namibia	9,170	29.11	0.19	0.08	20.37	8.73	0.42
Nepal	12,673	28.73	0.75	0.45	19.63	4.56	0.64
Nigeria	38,865	28.86	0.68	0.42	19.36	6.29	0.62
Peru	27,827	29.41	0.32	0.24	20.31	8.33	0.56
Sierra Leone	16,608	28.27	0.62	0.39	18.72	3.64	0.66
Togo	9,472	29.43	0.54	0.28	20.09	4.34	0.71

Notes: Panel A data come from the WORLD Policy Analysis Center. Panel B data come from female respondents ages 15–49 in study sample (Demographic and Health Surveys 2017).

a ban around the year 2003. Panel B indicates that most survey rounds were conducted toward the end of the period I examine (e.g., 2013), although a substantial group of respondents were surveyed closer to 2000. Panel C displays the distribution of years since the ban at the time of the survey round, revealing large groups

TABLE 2—SUMMARY STATISTICS—FEMALES, DHS SAMPLE

Sample:	Full sample			Urban			Rural		
	Mean	Standard deviation	Observations	Mean	Standard deviation	Observations	Mean	Standard deviation	Observations
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Age	29.62	9.60	254,764	29.66	9.60	108,131	29.60	9.60	146,633
Urban	0.42	0.49	254,764	1.00	0.00	108,131	0.00	0.00	146,633
<i>Marriage and fertility timing</i>									
Married	0.61	0.49	254,764	0.54	0.50	108,131	0.66	0.47	146,633
Age at first marriage	18.76	4.49	196,469	19.97	4.71	76,330	18.00	4.17	120,139
Married before age 18	0.34	0.47	254,764	0.23	0.42	108,131	0.42	0.49	146,633
Married before age 16	0.19	0.39	254,764	0.11	0.32	108,131	0.24	0.43	146,633
Married before age 14	0.06	0.24	254,764	0.03	0.18	108,131	0.08	0.27	146,633
Age first birth	19.93	4.17	190,085	20.84	4.38	74,821	19.34	3.92	115,264
<i>Schooling</i>									
Years of schooling	6.01	5.07	254,764	8.40	4.82	108,131	4.24	4.48	146,633
No education	0.30	0.46	254,764	0.15	0.35	108,131	0.41	0.49	146,633
Primary	0.25	0.43	254,764	0.19	0.39	108,131	0.29	0.46	146,633
Secondary	0.36	0.48	254,764	0.50	0.50	108,131	0.26	0.44	146,633
Higher	0.09	0.29	254,764	0.17	0.38	108,131	0.03	0.18	146,633
<i>Employment^a</i>									
Employed	0.54	0.50	254,764	0.50	0.50	108,131	0.57	0.50	146,633
Employed for cash	0.31	0.46	238,350	0.41	0.49	97,067	0.25	0.43	141,283
Employed for cash & in-kind	0.09	0.28	238,350	0.04	0.20	97,067	0.12	0.32	141,283
Employed for in-kind	0.03	0.18	238,350	0.01	0.10	97,067	0.05	0.22	141,283
Employed, not paid	0.12	0.33	238,350	0.06	0.24	97,067	0.16	0.37	141,283
Employed all year	0.38	0.48	241,280	0.40	0.49	98,609	0.36	0.48	142,671
Employed seasonally	0.13	0.33	241,280	0.07	0.25	98,609	0.17	0.37	142,671
Employed occasionally	0.05	0.22	241,280	0.05	0.21	98,609	0.05	0.22	142,671

Notes: Data come from female respondents age 15–49 in study sample Demographic and Health Surveys. Full set of employment questions not available for all countries. All indicator variables equal to one if true and zero otherwise.

^aThe Jordan DHS and the Kazakhstan DHS did not ask about form of compensation and the Jordan DHS did not ask about employment timing, resulting in smaller sample sizes.

of respondents interviewed in each of the years in the 10-year interval after a ban and a large mass of respondents interviewed just one year after a ban. I define an individual as being in a ban cohort (i.e., $post_{cj} = 1$) if they were under the age of 18 at the time of the ban.

Figure 3 presents the distribution of the subnational regional intensity measures.^{23,24} Table 3 presents summary statistics as well. For the “Main” measure—used in most regression specifications—in the full sample, the mean value is 1.10 and the median value is 0.99. For the “Alternative” measure—used as a robustness check—in the full sample, the mean value is 0.37 and the median value is 0.36. Both measures exhibit substantial variation and the correlation between the two measures is 0.97.²⁵ For both measures, the intensity is greater in rural areas than urban areas.

²³ The baseline specification uses women age 30 and below at the time of the ban to calculate the pre-ban intensity of child marriage. As sensitivity checks, I explore alternative age cut-offs for this calculation.

²⁴ There are 282 subnational regions (i.e., defined as the combination of DHS region and urban/rural) in my study sample. The median number in a country is 11. Almost all regions include both an urban and a rural area.

²⁵ Online Appendix Figure A10 presents the distribution of these measures country-by-country. The 17 panels in this figure reveal substantial subnational variation in the intensity of the ban.

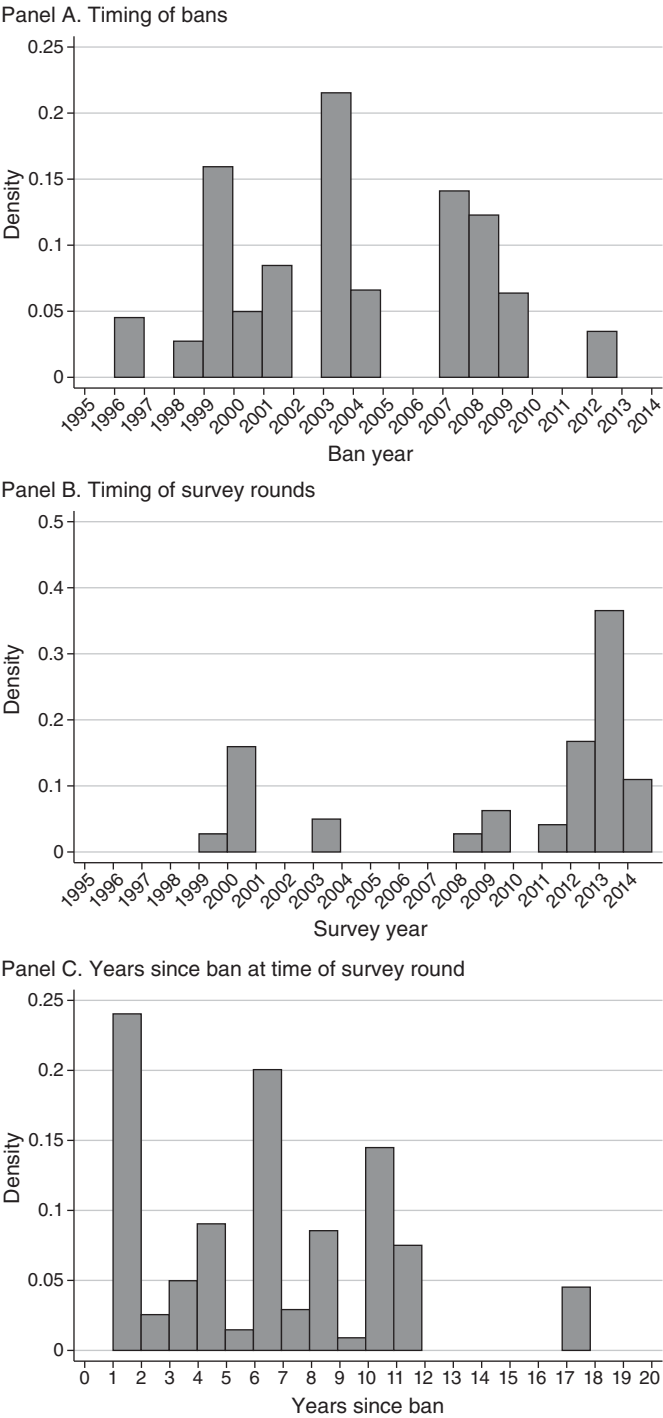


FIGURE 2. TIMING OF BANS AND SURVEY ROUNDS

Notes: Data in panel A are from the WORLD Policy Analysis Center database of minimum age at marriage laws spanning the period 1995–2012. Data in panel B are from the study sample DHS. Data in panel C are from the sources in panels A and B.

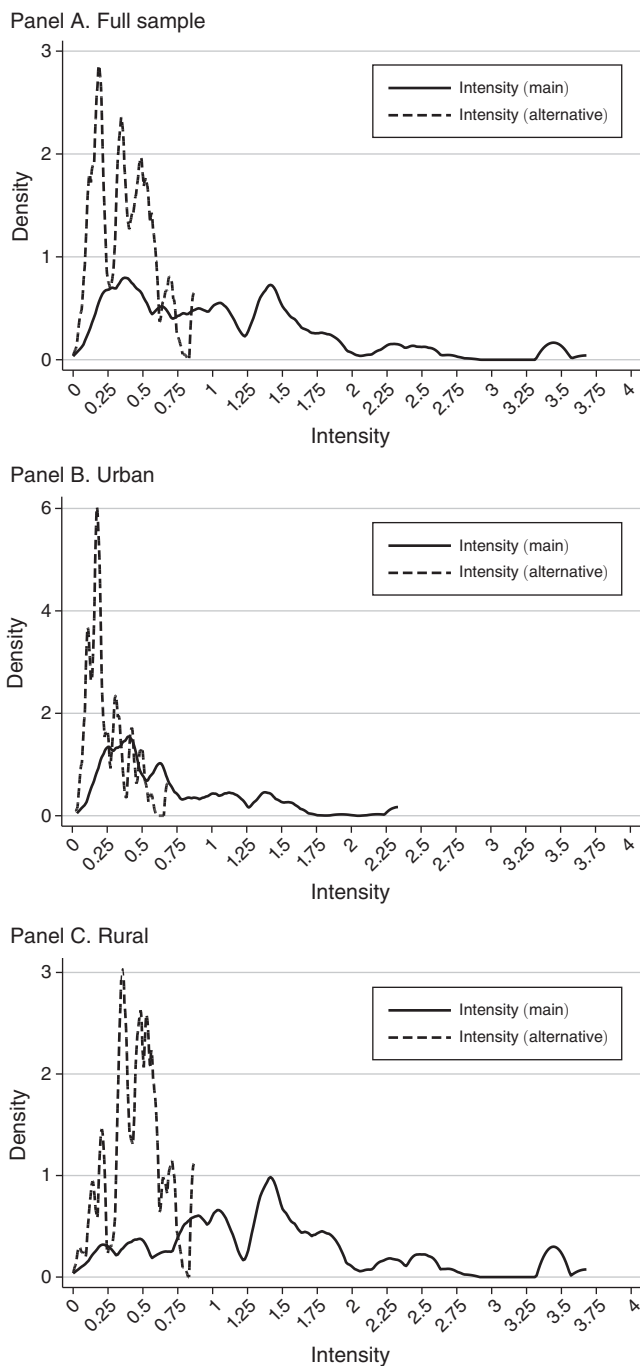


FIGURE 3. KERNEL DENSITY ESTIMATES OF DISTRIBUTION OF BAN INTENSITY

Notes: “Intensity (Main)” defined as the weighted average of the number of years an individual married before age 18, calculated using pre-ban cohorts. “Intensity (Alternative)” defined as proportion of individuals in pre-ban cohorts who married before age 18. Estimated using Epanechnikov kernel. Data come from female respondents age 15–49 in study sample DHS.

TABLE 3—SUMMARY STATISTICS FOR SUBNATIONAL REGIONAL INTENSITY MEASURES

Intensity measure: Sample:	Main			Alternative		
	Full sample (1)	Urban (2)	Rural (3)	Full sample (4)	Urban (5)	Rural (6)
Mean	1.10	0.70	1.40	0.37	0.25	0.46
Standard deviation	0.77	0.49	0.81	0.20	0.14	0.19
10th percentile	0.27	0.22	0.44	0.13	0.10	0.21
25th percentile	0.44	0.32	0.85	0.20	0.16	0.34
50th percentile	0.99	0.57	1.37	0.36	0.20	0.46
75th percentile	1.50	1.01	1.74	0.51	0.34	0.58
90th percentile	2.19	1.41	3.44	0.65	0.48	0.71
Minimum	0.00	0.03	0.00	0.00	0.02	0.00
Maximum	3.68	2.33	3.68	0.87	0.68	0.87
Observations	254,764	108,131	146,633	254,764	108,131	146,633

Notes: Data come from female respondents age 15–49 in study sample Demographic and Health Surveys. “Intensity (Main)” defined as the weighted average of the number of years an individual married before age 18, calculated using pre-ban cohorts. “Intensity (Alternative)” defined as proportion of individuals in pre-ban cohorts who married before age 18.

IV. Results

A. Effects of Child Marriage Bans

Table 4 presents the main results. The full study sample appears in panel A. The dependent variables in columns 1, 2, and 3 are an indicator variable for married before age 18, age at marriage in years, and age at first birth in years, respectively. The coefficient estimate in column 1 suggests that the bans did not reduce the likelihood of child marriage in the full sample.²⁶ The coefficient estimate in column 2 indicates that among ever married women a one-unit increase in the local intensity of the child marriage ban increased age at marriage by approximately 0.82 years (p -value < 0.01). To place this in context, it is useful to scale the effect size by one standard deviation: the coefficient estimate in column 2 suggests that a one standard deviation (i.e., 0.77 unit) increase in the local intensity of the ban was associated with an approximately 0.63 year increase in age at marriage.²⁷ The coefficient estimate in column 3 indicates that for women who have ever had a child, a one-unit increase in local ban intensity was associated with an approximately 0.54 year increase in age at first birth (p -value < 0.01); or, in other words, a one standard deviation increase in ban intensity was associated with an approximately 0.42 year increase in age at first birth.

²⁶Online Appendix Table A11 reports results for a closely related outcome—years married before age 18—finding an effect of the ban in the full sample: a reduction of 0.164 years in years married before age 18 (p -value < 0.01).

²⁷A one standard deviation increase in ban intensity (using the main ban intensity measure) corresponds to comparing a subnational region where all females married at age 18 or above (i.e., intensity equal to zero) to a subnational region where all females married prior to age 18 and the pre-ban average age at marriage was 17.33 (i.e., intensity equal to 0.77).

TABLE 4—EFFECT OF CHILD MARRIAGE BANS ON MARRIAGE, FERTILITY TIMING, SCHOOLING, AND EMPLOYMENT OUTCOMES

Dependent variable:	Married before age 18 (1)	Age at marriage (2)	Age at first birth (3)	Years of schooling (4)	Employed (5)
<i>Panel A. Full sample</i>					
Under age 18 at ban $\times$ region intensity	0.005 (0.013)	0.823 (0.089)	0.541 (0.072)	0.269 (0.196)	0.093 (0.009)
Sample mean of dependent variable	0.34	18.76	19.93	6.01	0.54
Observations	254,764	196,469	190,085	254,764	254,764
<i>Panel B. Urban females</i>					
Under age 18 at ban $\times$ region intensity	-0.092 (0.010)	1.347 (0.130)	1.049 (0.106)	1.064 (0.141)	0.053 (0.009)
Sample mean of dependent variable	0.23	19.97	20.84	8.40	0.50
Observations	108,131	76,330	74,821	108,131	108,131
<i>Panel C. Rural females</i>					
Under age 18 at ban $\times$ region intensity	0.018 (0.013)	0.676 (0.050)	0.379 (0.047)	-0.211 (0.100)	0.070 (0.016)
Sample mean of dependent variable	0.42	18.00	19.34	4.24	0.57
Observations	146,633	120,139	115,264	146,633	146,633
<i>Panel D. Countries with legal minimum of age 16 or 17 prior to ban</i>					
Under age 18 at ban $\times$ region intensity	-0.089 (0.012)	1.241 (0.116)	0.931 (0.085)	0.938 (0.171)	-0.118 (0.010)
Sample mean of dependent variable	0.32	19.10	20.39	6.24	0.46
Observations	74,648	61,314	58,176	74,648	74,648
<i>Panel E. Countries with legal minimum of age 14 or 15 prior to ban</i>					
Under age 18 at ban $\times$ region intensity	-0.017 (0.012)	0.975 (0.093)	0.709 (0.101)	0.807 (0.186)	0.101 (0.021)
Sample mean of dependent variable	0.32	18.95	19.94	6.11	0.55
Observations	117,607	87,353	86,185	117,607	117,607
<i>Panel F. Countries with no legal minimum prior to ban</i>					
Under age 18 at ban $\times$ region intensity	0.017 (0.015)	0.753 (0.015)	0.483 (0.015)	-0.034 (0.015)	0.089 (0.015)
Sample mean of dependent variable	0.41	17.99	19.31	5.55	0.61
Observations	62,509	47,802	45,724	62,509	62,509

Notes: Parameters estimated using ordinary least squares (OLS) regression. Robust standard errors clustered at sub-national region level are in parentheses. All specifications include indicator variables for subnational region of residence and for country-specific age in years.

Columns 4 and 5 present the results of the education and employment regressions. The point estimate in column 4 indicates that a one-unit increase in the local intensity of the ban was associated with a statistically insignificant increase of approximately 0.27 years of schooling (p -value = 0.172). The point estimate in column 5 indicates that a one-unit increase in the local intensity of the ban increased employment by 9.3 percentage points (p -value < 0.01). In other words, a one standard deviation increase in ban intensity was associated with an approximately 7.2 percentage point increase in the likelihood of employment.

Panels B and C repeat the analysis in panel A using the urban and rural subsamples, respectively. The results in panels B and C reveal larger effects of the bans on marriage age and age at first birth in urban areas than in rural areas. Notably, the results

suggest that the bans reduced child marriage for urban respondents. The coefficient estimate in column 1 of panel B indicates that a one-unit increase in ban intensity reduced the likelihood of child marriage by 9.2 percentage points (p -value < 0.01). Perhaps surprisingly, the coefficient estimate in column 4 of panel C suggests that the bans reduced years of schooling by 0.21 years (p -value < 0.05) for rural respondents, where the coefficient estimate in column 4 of panel B suggests that the bans increased years of schooling by approximately one year for urban respondents (p -value < 0.01). Lesser access to schooling may explain the fact that the bans did not increase years of schooling in rural areas, but seemingly would not explain the negative coefficient estimate.

The employment results in column 5 of panels B and C illuminate further explanation for this finding. The bans appear to have increased employment for urban and rural respondents, with a larger effect for rural respondents. Specialization in home production associated with household formation and/or increased economic burden of being a nonmarried adolescent in a high-child marriage prevalence environment may be greater in rural areas than in urban areas, leading to a larger employment effect and a muted, or negative, education effect. More generally, one reason the bans may have increased employment is that the bans increased years of schooling, which may have increased labor-market productivity or another dimension of employability. In rural areas and urban areas, the bans may have increased employment by postponing marriage and first birth, providing more latitude for employment.

Overall, the evidence presented thus far indicates that the bans increased age at marriage and age at first birth. Yet the bans appear to have only reduced child marriage in urban areas. A greater degree of imperfect enforcement in rural areas may explain this.

The results presented in panels D–F of Table 4 are consistent with imperfect enforcement in areas with weaker pre-ban restrictions on child marriage. These panels present results for three subsamples of countries: those with pre-ban legal minimums of “16 or 17,” of “14 or 15,” or no legal minimum, respectively. These results indicate that the bans had greater effects in countries with more restrictive pre-ban regimes. In contrast, a model of perfect enforcement would imply equality of effects of a one-unit increase in ban intensity in any of these three pre-ban regimes.

Table 5 examines the effects of the bans on the likelihoods of marrying before specific ages lower than age 18. If enforcement is imperfect, it may be enforced differentially by proximity to the new minimum legal age (i.e., age 18). The results in Table 5 are consistent with this hypothesis. In the full sample and in rural areas, the bans reduced the likelihood of marrying at younger ages at which pre-ban marriage was prevalent (i.e., age 13 or 14), but not age 18 per se (or even age 17). Similarly, countries with no legal minimum prior to the bans, and hence likely to have limited enforcement capacity, experienced the smallest changes in the likelihood of marrying before age 16 or age 17, yet exhibited substantial reductions in the likelihood of marrying at the youngest ages. Even if enforcement is imperfect, enforcement seems most likely to occur at the youngest ages. However, caution is warranted in overstating the imperfect enforcement interpretation as this analysis does not provide quasi-experimental evidence on the effect of varying enforcement. A different unobserved factor correlated with rural residence and with pre-ban regime may be

TABLE 5—EFFECTS OF CHILD MARRIAGE BANS ON MARRIAGE-AGE PROFILES

Dependent variable:	Married before age:						
	18 (1)	17 (2)	16 (3)	15 (4)	14 (5)	13 (6)	12 (7)
<i>Panel A. Full sample</i>							
Under age 18 at ban $\times$ region intensity	0.005 (0.013)	−0.007 (0.012)	−0.023 (0.009)	−0.029 (0.006)	−0.028 (0.005)	−0.017 (0.004)	−0.007 (0.003)
Sample mean of dependent variable	0.34	0.26	0.19	0.11	0.06	0.03	0.01
Observations				254,764			
<i>Panel B. Urban females</i>							
Under age 18 at ban $\times$ region intensity	−0.092 (0.010)	−0.092 (0.008)	−0.092 (0.009)	−0.069 (0.008)	−0.042 (0.004)	−0.020 (0.003)	−0.008 (0.003)
Sample mean of dependent variable	0.23	0.17	0.11	0.07	0.03	0.02	0.01
Observations				108,131			
<i>Panel C. Rural females</i>							
Under age 18 at ban $\times$ region intensity	0.018 (0.013)	0.003 (0.014)	−0.018 (0.011)	−0.032 (0.008)	−0.035 (0.008)	−0.022 (0.006)	−0.009 (0.005)
Sample mean of dependent variable	0.42	0.33	0.24	0.15	0.08	0.04	0.02
Observations				146,633			
<i>Panel D. Countries with legal minimum of age 16 or 17 prior to ban</i>							
Under age 18 at ban $\times$ region intensity	−0.007 (0.021)	−0.014 (0.020)	−0.026 (0.016)	−0.018 (0.011)	−0.008 (0.010)	−0.007 (0.006)	−0.005 (0.003)
Sample mean of dependent variable	0.32	0.23	0.16	0.09	0.05	0.02	0.01
Observations				74,648			
<i>Panel E. Countries with legal minimum of age 14 or 15 prior to ban</i>							
Under age 18 at ban $\times$ region intensity	−0.017 (0.012)	−0.025 (0.015)	−0.035 (0.015)	−0.033 (0.014)	−0.033 (0.014)	−0.024 (0.010)	−0.016 (0.007)
Sample mean of dependent variable	0.32	0.24	0.17	0.10	0.05	0.03	0.01
Observations				117,607			
<i>Panel F. Countries with no legal minimum prior to ban</i>							
Under age 18 at ban $\times$ region intensity	0.017 (0.015)	0.003 (0.013)	−0.017 (0.010)	−0.028 (0.005)	−0.028 (0.005)	−0.015 (0.004)	−0.004 (0.002)
Sample mean of dependent variable	0.41	0.33	0.26	0.16	0.09	0.04	0.02
Observations				62,509			

Notes: Regression sample in columns 1–7 is females age 15–49. Regression sample in column 8 is “ever married” females age 18–49. Parameters estimated using ordinary least squares (OLS) regression. Robust standard errors clustered at subnational region level are in parentheses. All specifications include indicator variables for subnational region of residence.

driving this heterogeneous response. Another caveat about the results in Table 5 is that the estimated effects of the bans on the likelihood of marrying before specific ages below the age of 18 do not readily fit the effect of the bans on the mean age at marriage. One possible explanation—also consistent with the evidence that the bans increased tertiary education—is that the bans indirectly affected marriage markets and educational trajectories for women age 18 and above.

B. Robustness Checks

Online Appendix Table A13 presents sensitivity to the age cutoff for control respondents. Panel A limits the regression sample to respondents under the age of

40 at the time of the ban. Panels B–E repeat this exercise with increasingly lower age cutoffs. Panels F and G are similar, yet use females between the ages of 25–30 and 30–40, respectively, as control respondents. Columns 1–5 display results for the full sample. Columns 6–10 display results for the urban sample. Columns 11–15 display results for the rural sample. Throughout, the results are very similar to those in the baseline specification (i.e., Table 4). An exception to this pattern is that the point estimate in the “Married before age 18” regression is negative (i.e., -0.022) and statistically significant (p -value < 0.05) when limiting control respondents to those age 25 and below or to age 20 and below.

Online Appendix Table A14 repeats the spirit of the sensitivity analysis in the previous table by adjusting the age cutoff for treated respondents. Panel A limits the treated respondents in the regression sample to those age 0 or above at the time of the ban. Subsequent panels examine increasingly higher age cutoffs. A general pattern emerges from this analysis: as the age cutoff for treated respondents increases, the estimated effects become attenuated for most outcomes. This pattern is consistent with respondents who were older at the time of the ban yet still under the age of 18 having less exposure to the ban on the intensive margin and hence demonstrating smaller behavior change. Notably, the point estimate in column 1—the “Married before age 18” regressions for the full sample—becomes positive (i.e., approximately 0.02) and statistically significant (p -values < 0.05) when limiting treated respondents to age 14 and above (or age 15 and above). This may reflect an increase in child marriage right near the new legal minimum in anticipation of future increased enforcement.

C. Event Study Analyses

Figure 4 displays the coefficient estimates from event study analyses of the effect of the bans on “Married before age 18.” Panels A, B, and C present the results for the full sample, urban sample, and rural sample, respectively. Formally, I estimate the following semi-parametric specification:

$$(4) \quad outcome_{icrj} = \sum_{k=4-}^{46+} \beta_k 1(\tau_t = k) Xintensity_{rj} + \gamma_{cj} + \delta_{rj} + \varepsilon_{icrj},$$

where τ_t indicates the respondent’s age at the time of the ban in the data and the remaining notation follows that in equation (1).²⁸ The x -axis measures age at the time of the ban and the y -axis measures the “effect size” of the ban. For ages 18 and above at the time of the ban, the ban should not have directly affected marriage

²⁸ The event study analyses pool individuals age four and below at the time of the ban (approximately 1 percent of the full sample). They also pool individuals age 46 and above at the time of the ban (approximately 1 percent of the full sample). The sample sizes for these age groups fall below 200 and, for individuals age 4 or less, only come from one country (i.e., Namibia), leading to much larger standard errors than for other ages and generally noisy point estimates. Including (or not including) these respondents does not affect the observed pattern of zero “effects” for individuals age 18 and above at the time of the ban, a break in the pattern around approximately age 17, and negative (or, in the case of the schooling/employment results, positive) point estimates for individuals age 17 and below at the time of the ban.

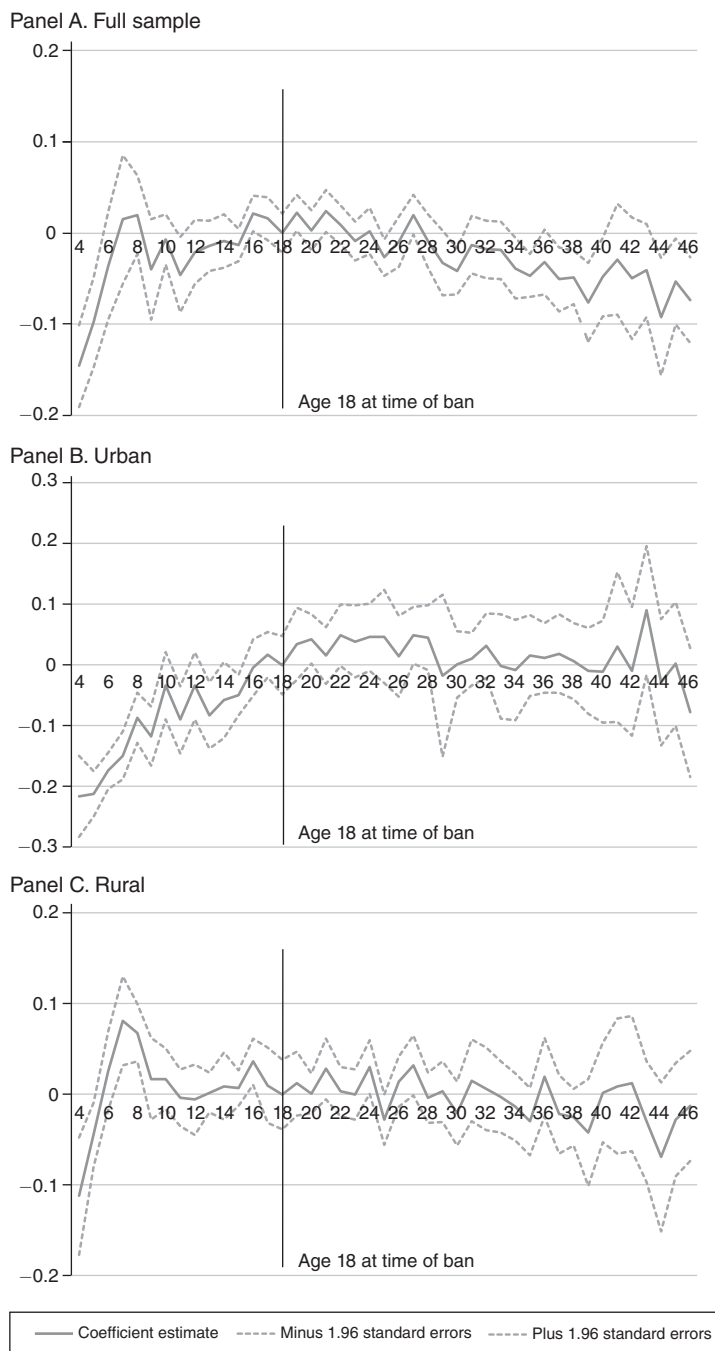


FIGURE 4. EVENT STUDY ANALYSIS, MARRIED BEFORE AGE 18

Notes: The solid line plots the point estimates from an event study regression where the dependent variable is an indicator equal to 1 if the respondent married before age 18. The dashed lines are the associated 95 percent confidence intervals.

outcomes because the ban did not apply to these age cohorts.²⁹ Consistent with this view, the point estimates for these ages hover around zero and are not statistically significant. For cohorts age 17 and below in urban areas, the ban began to reduce the likelihood of child marriage, with larger effects (and a greater likelihood of statistical significance) at younger ages (i.e., for individuals who were less likely to already be married at the time of the ban). In rural areas, the event study analysis of “Married before age 18” displays little to no evidence of a break around age 17.

Figures 5 and 6 repeat this analysis using age at marriage and age at first birth as dependent variables, respectively. Figures 7 and 8 repeat this analysis using years of schooling and employment as dependent variables, respectively. Aside from the rural education figure (i.e., Figure 7 panel C, which displays a nearly flat zero “effect” throughout virtually all ages), the figures reveal a pattern of a consistently estimated zero “effect” for cohorts that should not have been directly affected by the ban, a sharp change in this pattern at or around age 18, and increasingly larger effects for younger cohorts below the age of 18.³⁰ The point estimates for younger cohorts in the employment regressions are often not statistically significant at a single age, yet follow the same pattern as the marriage and schooling event study analyses, and the parameterized difference-in-differences analysis (i.e., Table 4) indicates that the bans increased employment.

One explanation for large effect sizes at younger ages is that married females below the age of 18 at the time of the ban were “grandfathered” into the previous legal regime. Although there appears to be limited evidence on this practice, it is as plausible as a country annulling these marriages at the time of the ban.³¹ Another explanation is that a longer time horizon allowed younger respondents and their households greater latitude to make household formation decisions compliant with the new legal regime.

D. *Instrumental Variables Results*

Several economic studies have measured the effect of age at marriage on schooling (e.g., Ambrus and Field 2008, Hicks and Hicks 2015). Table 6 presents instrumental variables estimates of the effect of age at marriage on educational attainment. I instrument for age at marriage with child marriage bans using the difference-in-differences specification in equation (1) as my first-stage specification.

²⁹ Note that the “treated” respondents are those age 17 and less at the time of the ban and that these respondents appear to the left of the vertical break line.

³⁰ One interpretation of Figure 8 is that it includes an underlying differential time trend correlated with the pre-ban intensity measure. To further unpack this hypothesis, online Appendix Figures A3–A7 (and Figure A9) present the event study parameters from specifications using the regional trends included in online Appendix Table A6, panel D. For years of schooling (i.e., online Appendix Figure A6), the figure clearly reveals a break at age 18. Combined with the results of the robustness test in online Appendix Table A6, panel D, this provides reassurance that the results do not reflect differential subnational region time trends. For the other figures, the event study analyses with and without subnational region time trends look very similar, again consistent with the robustness test in online Appendix Table A6, panel D.

³¹ In a review of age at marriage laws in Africa, Odala (2013) reports 11 countries that “ban or invalidate marriages below the minimum age”: Angola, Burundi, Cape Verde, The Gambia, Mauritius, Mozambique, Namibia, Sao Tome and Principe, South Africa, Tanzania, and Uganda.

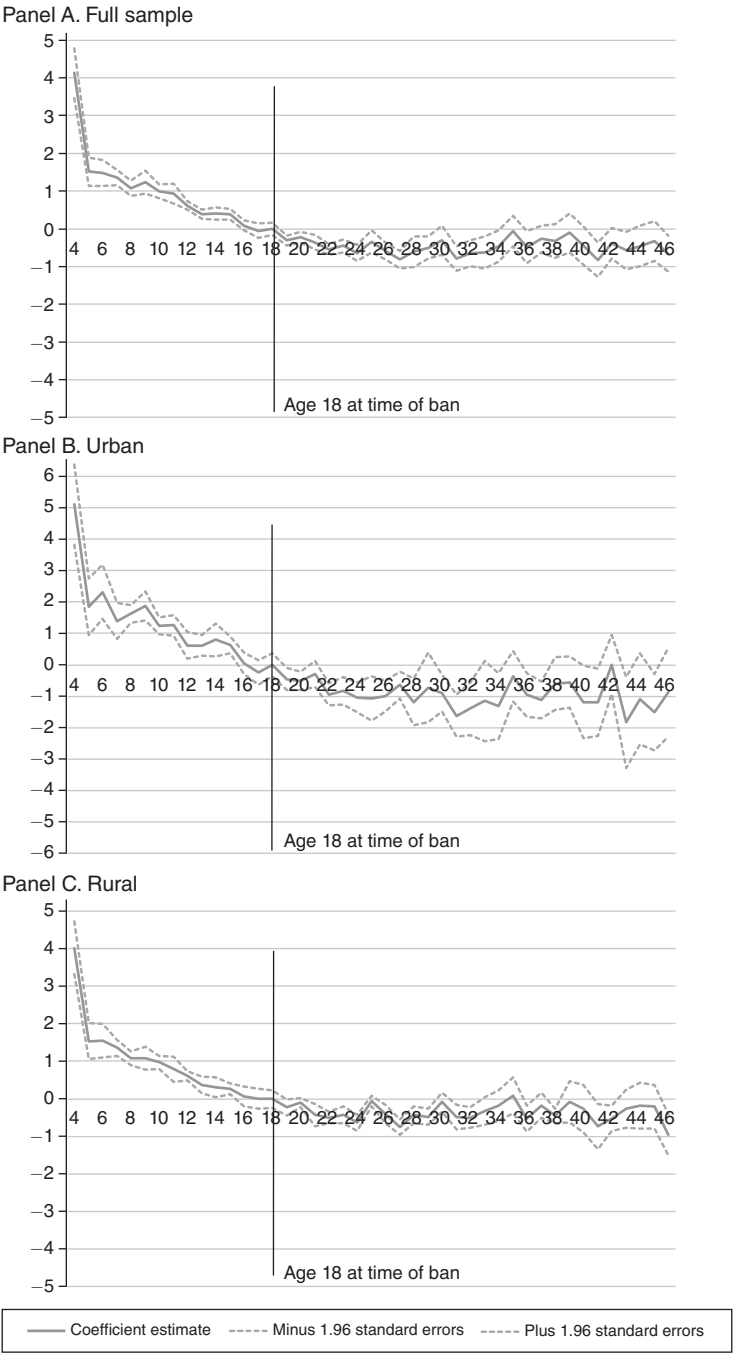


FIGURE 5. EVENT STUDY ANALYSIS, AGE AT MARRIAGE

Notes: The solid line plots the point estimates from an event study regression where the dependent variable is an indicator equal to 1 if the respondent married before age 18. The dashed lines are the associated 95 percent confidence intervals.

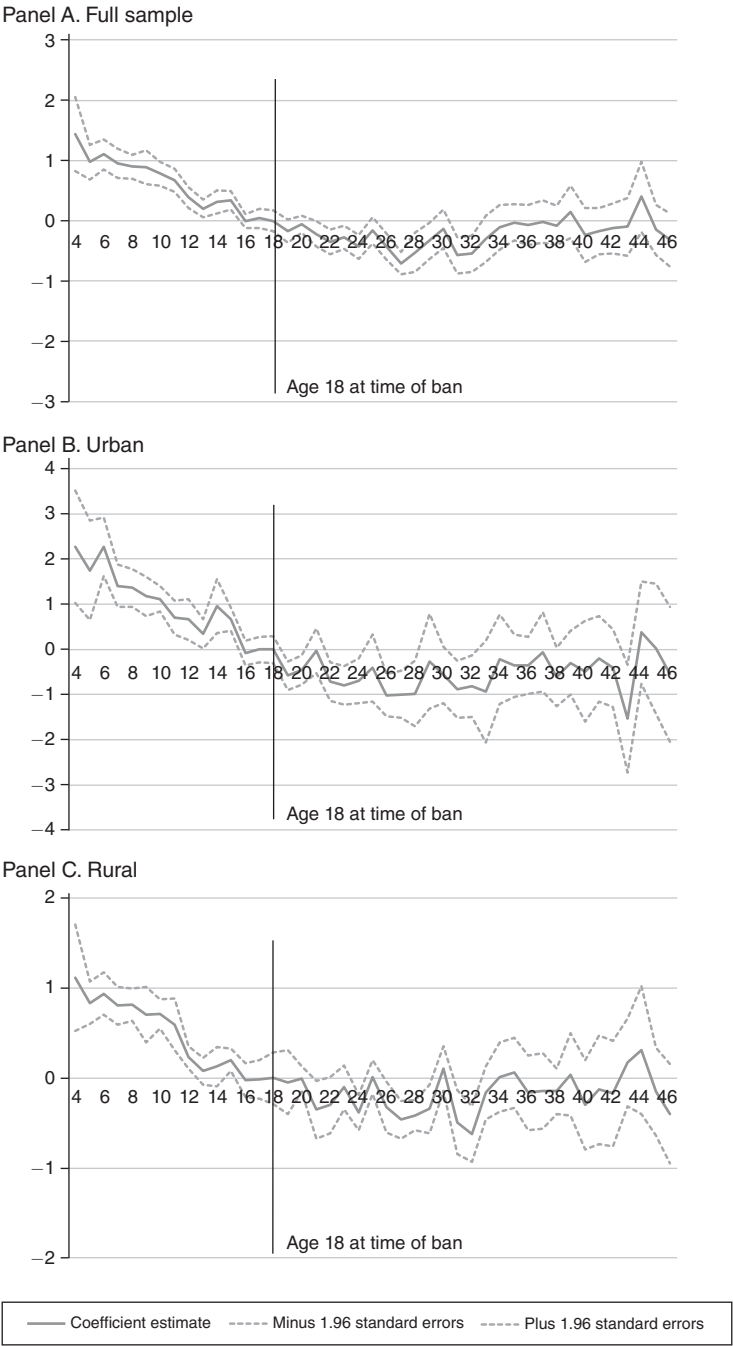


FIGURE 6. EVENT STUDY ANALYSIS, AGE AT FIRST BIRTH

Notes: The solid line plots the point estimates from an event study regression where the dependent variable is an indicator equal to 1 if the respondent married before age 18. The dashed lines are the associated 95 percent confidence intervals.

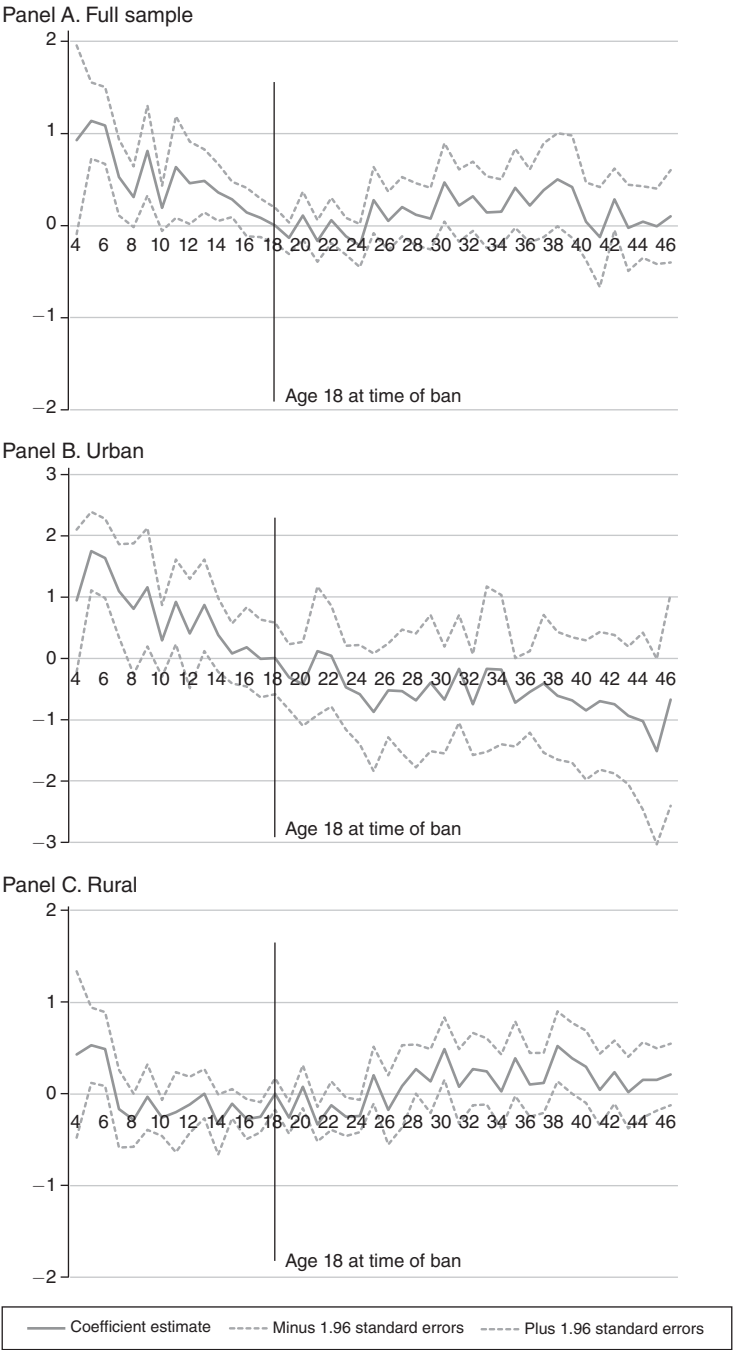


FIGURE 7. EVENT STUDY ANALYSIS, YEARS OF SCHOOLING

Notes: The solid line plots the point estimates from an event study regression where the dependent variable is an indicator equal to 1 if the respondent married before age 18. The dashed lines are the associated 95 percent confidence intervals.

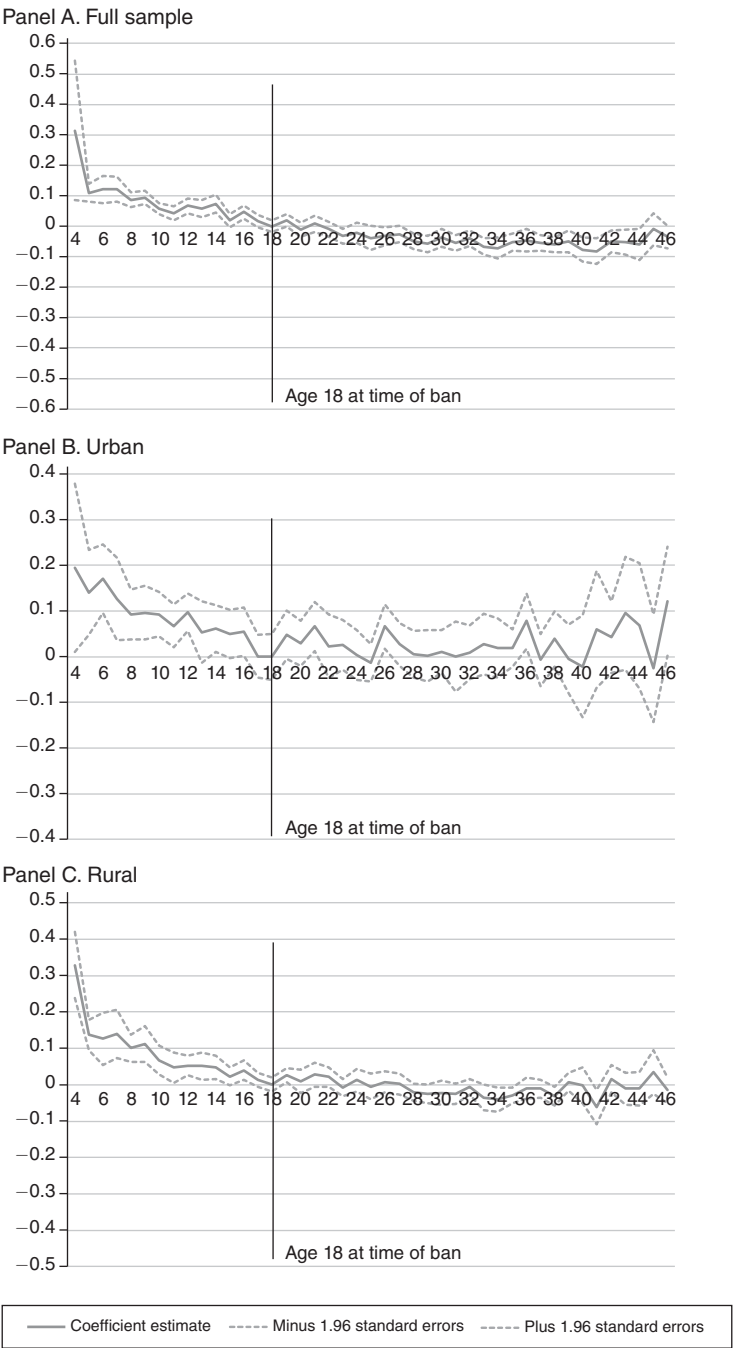


FIGURE 8. EVENT STUDY ANALYSIS, EMPLOYED

Notes: The solid line plots the point estimates from an event study regression where the dependent variable is an indicator equal to 1 if the respondent married before age 18. The dashed lines are the associated 95 percent confidence intervals.

TABLE 6—INSTRUMENTAL VARIABLES ESTIMATES OF EFFECT OF AGE AT MARRIAGE ON EDUCATIONAL ATTAINMENT

Dependent variable: Sample:	Years of schooling		
	Full sample (1)	Urban (2)	Rural (3)
Age at marriage	0.174 (0.149)	0.432 (0.118)	−0.223 (0.169)
Sample mean of dependent variable	5.372	7.937	3.742
First-stage <i>F</i> -stat	85.20	107.76	180.46
Observations	196,469	76,330	120,139

Notes: Parameters estimated using two-stage least squares (2SLS). Robust standard errors clustered at subnational region level are in parentheses. All specifications include indicator variables for subnational region of residence and country-specific age in years.

Column 1 reports the results for the full sample. Columns 2 and 3 report results for urban and rural respondents, respectively. The point estimate in column 2 indicates that among urban respondents, postponing marriage by one year increased educational attainment by 0.432 years (p -value < 0.01). In contrast, the point estimate in column 3 is negative, although not statistically significant.

Caution should be used in interpreting the instrumental variables results. Social/peer effects or concurrent policies may threaten the exclusion restriction. For example, a child marriage ban may have affected the returns to schooling for some respondents even if the ban did not directly affect their age at marriage. Concurrent policies could include strengthening women's private property rights (e.g., Harari 2014, La Ferrera 2014, Hallward-Dreimeier 2015), right to work outside of the home (e.g., Hallward-Dreimeier 2015), or right to control their own bank account (e.g., World Bank 2015). Although I demonstrate in the reduced-form analysis that a main potential concurrent policy—compulsory schooling laws—is not driving the main results, the exclusion restriction in the IV analysis is sufficiently restrictive that it may be unlikely to be satisfied. Thus, there is concern that the IV results are not strictly identifying the causal effect of postponing marriage on educational attainment.

V. Conclusion

Worldwide, among women age 20–24 more than 1 out of 10 women married before age 15 and more than 1 out of every 3 women married before the age of 18 (UNICEF 2014). Early marriage is associated with poor socioeconomic and female-empowerment outcomes among women and low overall levels of economic development and growth, yet there is little research on a common—albeit not universal—policy: child marriage bans. I measure the effects of child marriage bans on female schooling and labor market outcomes in a sample of 17 low- and middle-income countries. My results using the full study sample suggest that raising the minimum legal age of marriage to 18 increased age at marriage, age at first birth, and the likelihood of employment. In urban areas, the bans also reduced child marriage and increased educational attainment.

Policymakers should consider increasing the use and enforcement of child marriage bans as a means of increasing economic development. As of 2012, more than 50 countries in the world—not just low- and middle-income countries, as in my study sample, but also large, high-income countries including the United States, the United Kingdom, Japan, and France—allowed child marriage. Not only would this step address what many see as a human rights issue (United Nations 1962), it may also promote economic development by allowing girls to go to school and by promoting female employment.

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